

JAY CHAFEKAR

Graduate Full-Stack Developer | Python · Django · React · SQL

+44 7774 939653 • jaychafekar312003@gmail.com • London, UK • linkedin.com/in/jay-chafekar • github.com/Jaychafekar

PROFESSIONAL SUMMARY

Final-year BSc Computer Science student at the University of Westminster, graduating July 2026 with a predicted 2:1, applying for graduate full-stack roles. I've shipped full-stack web apps across a deepfake-detection final-year project (FastAPI + React + Docker), a Sky-sponsored Django + Bootstrap app with role-based access, and a backend internship at Technokraft. Comfortable across REST API design, OOP, server-rendered and SPA frontends, testing, and Agile.

TECHNICAL SKILLS

Languages: Java, Python, JavaScript (ES6+), SQL, C++, C

Backend: Django, FastAPI, JAX-RS, REST APIs, Object-Oriented Programming (OOP), MVC, microservice patterns

Frontend: React, Vite, HTML5, CSS3, Bootstrap

Databases & Testing: MySQL, PostgreSQL, SQLite, MongoDB; JUnit, pytest, unit & integration testing, debugging

DevOps & Tools: Git/GitHub, Docker, Docker Compose, Linux, CI/CD, Maven, Swagger, Postman

Practices: Agile/Scrum, code reviews, pair programming, TDD, technical documentation

CS Fundamentals: Data structures, BFS/DFS, Edmonds-Karp, minimax/alpha-beta; TensorFlow

PROJECTS

PixelProof — Deepfake Detection & Verification Platform — FastAPI · TensorFlow · React · Docker *Final Year Project, 2025 – 26*

- Built the deepfake-detection stack as four Docker Compose services — FastAPI inference API, fine-tuned MobileNetV2 model, React/Vite review UI, SQLite store — with pytest covering signing, verification, and inference-error paths.
- Reached 69.65% cross-dataset accuracy and 0.7772 ROC-AUC on a held-out corpus, reporting those over the same-dataset numbers that usually look better than the model really is.
- Switched from a CNN-LSTM to MobileNetV2 when the temporal model kept overfitting; every verdict comes back signed with HMAC-SHA256 over image hash, verdict, model version, and timestamp — a tamper-evident audit trail.

Sky × University of Westminster — Industry Collaboration · Django · Bootstrap · SQL *Jan – Apr 2025*

- A 3-person team project building a Squad Health Check web app for Sky, with their engineers running code reviews and giving feedback throughout the module.
- Built the Django backend: a custom User model extending AbstractUser with five roles (employee, team lead, department lead, senior manager, admin), role-based access on every view, and a domain model across Departments, Teams, Sessions, Health Cards, and Votes.
- Wrote views, forms, and server-rendered Bootstrap templates for each role's dashboard, plus management commands for permission setup.

Bookstore REST API — Java · JAX-RS · SQL · Swagger *2025*

- A layered Java REST service on JAX-RS — thin resource classes calling into a service layer, with persistence behind that — so the business logic stayed unit-testable in isolation under JUnit.
- Books and customers modelled as related resources, with idempotent PUT/DELETE and POST for creation; proper HTTP status codes (201/204/404/409) rather than 200-for-everything.
- SQL persistence with indexed lookups on ISBN and customer ID; JAX-RS annotations generate a Swagger/OpenAPI spec that stays in sync with the code.

Max Flow Network — Java · Graph Algorithms *2025*

- Implemented Edmonds-Karp on a residual-graph representation, scaling to 1,000+ nodes — using BFS for augmenting paths gives $O(VE^2)$ and avoids the pathological worst cases of plain Ford-Fulkerson.
- Picked an adjacency-list representation over an adjacency matrix to keep memory linear in edges, which mattered once node counts crossed several hundred.
- A small test harness checks flow conservation and edge feasibility on every iteration, with diagnostic dumps so a failure points to a specific augmenting step.

EXPERIENCE

London Success Academy — Software Engineering Intern, London (Remote) *May 2026 – Present*

- Building a full-stack web app in Python and JavaScript covering REST APIs, user authentication, and database integration.
- Working through weekly code reviews with mentors, pairing Git workflow and Agile practices with testing on what I push to GitHub.

Technokraft Ltd — Backend Developer Intern, Nashik *Jun – Aug 2023*

- Wrote internal REST APIs for an IT incident reporting workflow used by internal teams.
- Worked on operational dashboards that pulled from those APIs, aimed mainly at non-technical users; tuned SQL on the slower reporting endpoints.

EDUCATION

University of Westminster, London — BSc (Hons) Computer Science *Sep 2023 – Jul 2026 | Predicted 2:1*